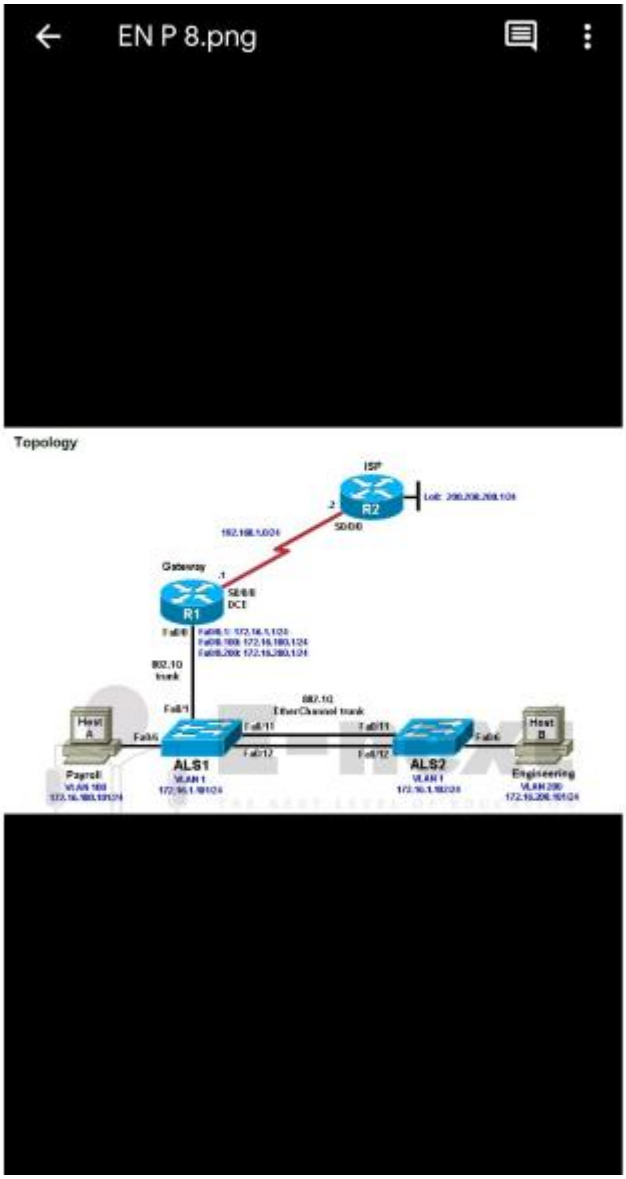
P 8 Inter-VLAN Routing with an External Router



Background

Inter-VLAN routing using an external router can be a cost-effective solution when it is necessary to segment a

network into multiple broadcast domains. In this lab, you split an existing network into two separate VLANs on

the access layer switches, and use an external router to route between the VLANs.

Required Resources

• 2 routers (Cisco 1841 with Cisco IOS Release 12.4(24)T1 Advanced IP Services or comparable)

• 2 switches (Cisco 2960 with the Cisco IOS Release 12.2(46)SE C2960-LANBASEK9-M image or comparable)

Router 1 – ISP

```
enable
configure terminal
hostname ISP

interface Loopback0
ip address 200.200.200.1 255.255.255.0
no shutdown
exit

interface Serial0/0/0
ip address 192.168.1.2 255.255.255.0
no shutdown
exit

ip route 172.16.0.0 255.255.0.0 192.168.1.1
end
```

Router 2 – Gateway

```
enable
configure terminal
hostname Gateway

interface Serial0/0/0
ip address 192.168.1.1 255.255.255.0
clock rate 64000
no shutdown
exit

ip route 0.0.0.0 0.0.0.0 192.168.1.2
end
```

Switch 1 – ALS1

```
enable
configure terminal
hostname ALS1

interface vlan 1
ip address 172.16.1.101 255.255.255.0
no shutdown
exit

ip default-gateway 172.16.1.1
```

```
enable secret cisco

line vty 0 15
password cisco
login
exit
```

Switch 2 – ALS2

```
enable
configure terminal
hostname ALS2

interface vlan 1
ip address 172.16.1.102 255.255.255.0
no shutdown
exit

ip default-gateway 172.16.1.1

enable secret cisco

line vty 0 15
password cisco
login
exit
```

Trunk + EtherChannel

ALS1

```
configure terminal
interface range fastEthernet 0/11 - 12
switchport mode trunk
channel-group 1 mode desirable
end
```

ALS2

```
configure terminal
interface range fastEthernet 0/11 - 12
switchport mode trunk
channel-group 1 mode desirable
end
```

VTP Configuration

ALS2

```
configure terminal
vtp mode client
end
```

ALS1

```
configure terminal
vtp domain SWLAB
vtp version 2

vlan 100
name Payroll
exit

vlan 200
name Engineering
exit
```

Access Ports

ALS1

```
configure terminal
interface fastEthernet 0/6
switchport mode access
switchport access vlan 100
spanning-tree portfast
exit
```

ALS2

```
configure terminal
interface fastEthernet 0/6
switchport mode access
switchport access vlan 200
spanning-tree portfast
exit
```

Trunk to Router (ALS1)

```
configure terminal
interface fastEthernet 0/1
switchport mode trunk
end
```

Gateway Router – Inter-VLAN Routing (Router-on-a-Stick)

```
enable
configure terminal

interface fastEthernet 0/0
no shutdown
exit

interface fastEthernet 0/0.1
description Management VLAN 1
encapsulation dot1Q 1 native
ip address 172.16.1.1 255.255.255.0
exit

interface fastEthernet 0/0.100
description Payroll VLAN 100
encapsulation dot1Q 100
ip address 172.16.100.1 255.255.255.0
exit

interface fastEthernet 0/0.200
description Engineering VLAN 200
encapsulation dot1Q 200
ip address 172.16.200.1 255.255.255.0
exit

end
```

Verification Commands (run separately)

```
show vlan brief
show vtp status
show etherchannel summary
show ip interface brief
show interfaces description
```